

Two maximum antichains disprove conjecture 00000008544

Local AI-assisted solution draft
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Claim being disproved. The conjecture claims that a Cartesian product of total-order chains has a unique maximum antichain. The product of three two-element chains is a counterexample.

Construction. Let $P = \{0, 1\}^3$ with the coordinatewise order, equivalently the Boolean lattice of subsets of $\{1, 2, 3\}$. Define

$$\mathcal{A} = \{100, 010, 001\}, \quad \mathcal{B} = \{110, 101, 011\}.$$

Distinct members of either family are incomparable: they have equal Hamming weight, and a coordinatewise inequality at equal weight forces equality. Both families have size three and they are distinct.

Maximum size. For completeness, partition all eight elements into the following three chains:

$$000 < 001 < 011 < 111, \quad 010 < 110, \quad 100 < 101.$$

Every antichain meets each chain at most once, so it has at most three elements. Consequently both $\mathcal{A}$ and $\mathcal{B}$ are maximum antichains. They are the two middle rank layers, demonstrating why a uniqueness claim fails in this example. The Sperner number is 3; its precise expression elsewhere in the conjecture need not be settled to refute uniqueness.

Formal verification and semantic correspondence. The accompanying `Main.lean` uses Lean 4.19.0 Std. The eight elements are indexed by `Fin 8`; `cube` and `unCube` are explicitly checked inverse maps with `(Fin 2)3`. The order `Below` is exactly the coordinatewise order under this identification, and totality of the factors and the resulting partial-order laws are checked.

Subsets are represented by eight-bit masks `Fin 256`. Membership is defined by the corresponding bit; `every_subset_encoded` proves that every Boolean membership function on the eight elements has such an encoding, so the subset enumeration is exhaustive. `Antichain` uses actual membership and comparability, and `cardinality` counts actual members. The theorem `sharp_upper_bound` checks all 256 subsets and proves that every antichain has at most three elements. This is an independent finite verification of the chain-partition argument above.

The masks 22 and 104 represent $\mathcal{A}$ and $\mathcal{B}$ respectively; their member descriptions, sizes, maximum status, and inequality are all checked. The final theorem `conjecture8544_false` directly negates uniqueness of a maximum antichain on this chain product. It does not merely exhibit incomparable elements without checking maximal size.

All finite checks use kernel reduction via `decide`. The final disproof has only the standard axiom `propext`; the general encoding lemma additionally uses standard `Quot.sound` through function extensionality. There are no placeholders, `native_decide`, or additional axioms.

Reproduction. Run `lean Main.lean` with Lean 4.19.0.